Comprehensive Analytical Solutions: Electrical Instrumentation

Semester Examination – Part I

Contents

QUESTION 1: JUSTIFICATIONS AND CORRECTIONS

1(a) LVDT Phase Compensation Network

Statement: "In an LVDT, a leading phase angle at the output can be retarded by connecting a C-R high pass filter network at the output."

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Explanation: An ideal Linear Variable Differential Transformer (LVDT) exhibits a phase shift between its primary excitation voltage and its secondary differential output voltage. Let the primary coil possess a resistance R_p and an inductance L_p. The primary current I_p, and consequently the magnetic flux Φ, lags the excitation voltage E_{in} by an angle:

$$\phi_{primary} = \tan^{-1}\left(\frac{\omega L_p}{R_p}\right) \tag{1}$$

The induced secondary voltage depends on the rate of change of flux ($d\Phi/dt$), introducing an inherent 90° phase lead. The net phase angle of the output voltage E_{out} relative to E_{in} is often a leading angle if the secondary load is highly resistive.

To "retard" or correct a leading phase angle, we must introduce a phase *lag*. A cascaded C-R high-pass filter (capacitor in series, resistor in shunt) has the following transfer function:

$$H(j\omega) = \frac{j\omega CR}{1 + j\omega CR} \tag{2}$$

The phase angle introduced by this high-pass filter is:

$$\theta_{HPF} = 90^\circ - \tan^{-1}(\omega RC) \tag{3}$$

This angle is strictly positive (a phase **lead**). Adding a lead to an already leading LVDT output completely ruins the compensation.

To correctly retard the phase to zero, an **R-C low-pass filter** (resistor in series, capacitor in shunt) must be employed. Its transfer function is:

$$H(j\omega) = \frac{1}{1 + j\omega RC} \tag{4}$$

Yielding a phase angle of:

$$\theta_{LPF} = -\tan^{-1}(\omega RC) \tag{5}$$

This negative angle (lag) effectively cancels the inherent leading angle of the LVDT.

Orthogonal Block Diagram of Correct Scheme:

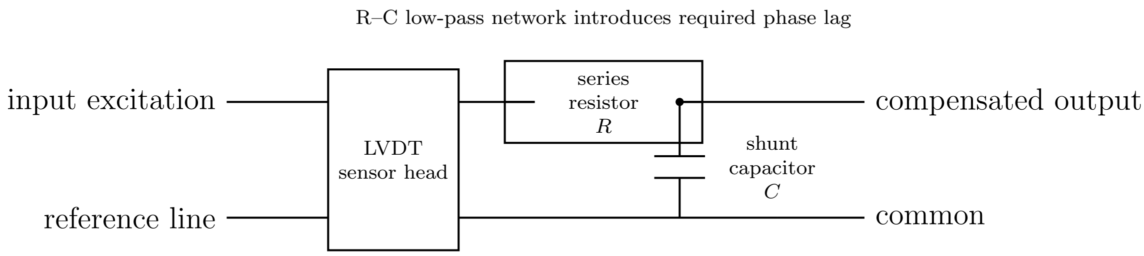

Figure 1: Correct R–C low-pass phase-compensation network for LVDT output.

1(b) Piezoelectric Bandwidth and Time Constant

Statement: ”The bandwidth of a piezoelectric displacement transducer system can be expanded by choosing a lower time constant.”

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Explanation: A piezoelectric transducer produces an electrical charge q proportional to the applied dynamic force. Electrically, the bare crystal acts as a charge generator in parallel with an internal crystal capacitance C_p and a very high internal leakage resistance R_p. When connected to an amplifier with input resistance R_a and cable capacitance C_c, the entire equivalent circuit simplifies to a first-order high-pass filter.

The total equivalent resistance is $R_{eq} = \frac{R_p R_a}{R_p + R_a}$. The total equivalent capacitance is $C_{eq} = C_p + C_c + C_a$. The governing time constant of the discharge circuit is:

$$\tau = R_{eq} C_{eq} \tag{6}$$

The transfer function relating output voltage $E_0(s)$ to input displacement $X_i(s)$ takes the strict high-pass form:

$$\frac{E_0(s)}{X_i(s)} = K_{sensitivity} \left[\frac{s\tau}{1 + s\tau} \right] \tag{7}$$

In the frequency domain ($s = j\omega$), the magnitude is:

$$|G(j\omega)| = K_{sensitivity} \frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}} \tag{8}$$

The usable, flat bandwidth of a piezoelectric sensor exists at frequencies significantly higher than the lower cut-off frequency (ω_L). The lower cut-off frequency (the -3 dB point) is strictly defined by the time constant:

$$\omega_L = \frac{1}{\tau} \tag{9}$$

If an engineer selects a **lower** time constant ($\tau \downarrow$), the lower cut-off frequency ω_L shifts proportionally **higher** ($\omega_L \uparrow$). This means the sensor becomes blind to low-frequency vibrations, thereby **contracting** (reducing) the usable bandwidth, not expanding it. To measure very slow, low-frequency dynamic phenomena, one must artificially create a massive, highly elevated time constant (typically via a specialized Charge Amplifier).

Orthogonal Diagram of Equivalent Circuit:

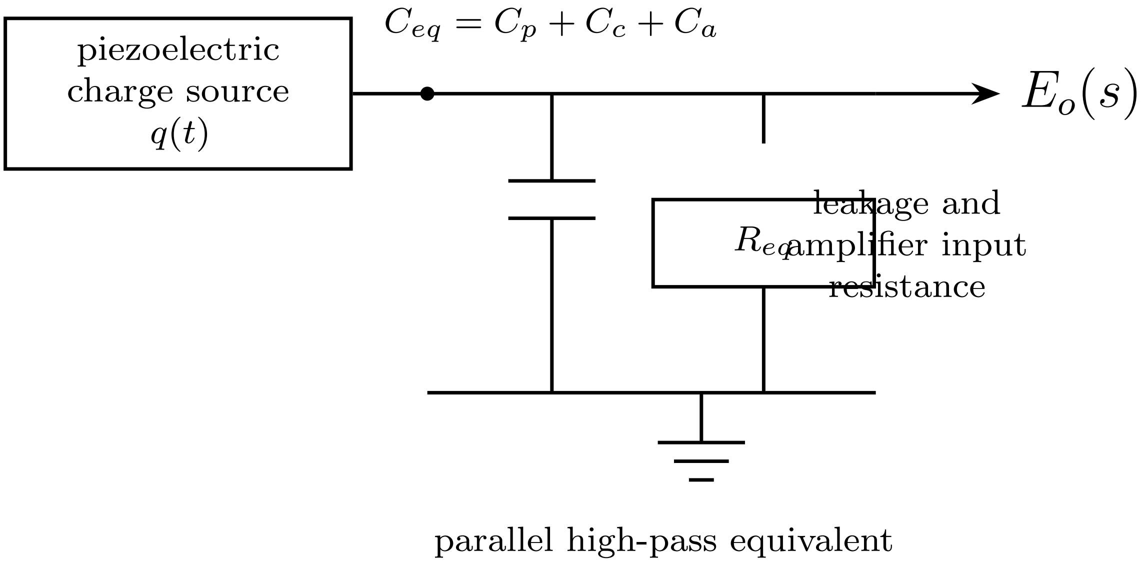

Figure 2: Clean equivalent circuit of a piezoelectric transducer system.

1(c) Angular Capacitive Transducer Sensitivity

Statement: "When a capacitive transducer is employed for measurement of angular displacement using the principle of change in overlapping area, the sensitivity will be directly proportional to the displacement between the plates and inversely proportional to the radius of the smaller plate."

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Explanation: Consider a variable-area capacitive angular transducer consisting of two parallel semi-circular plates. One plate is absolutely fixed (stator), and the other plate rotates on a central shaft (rotor). The overlapping geometry forms the active area of the capacitor.

Let the uniform gap distance between the plates be strictly held at d. Let the effective radius of the overlapping sector be r. When the rotor undergoes an angular displacement θ (measured in radians), the overlapping sector area A is precisely calculated by the geometric formula:

$$A(\theta) = \frac{1}{2}r^2\theta \tag{10}$$

Substituting this area into the fundamental parallel-plate capacitance equation $C = \frac{\epsilon A}{d}$, we obtain the transducer's transfer function:

$$C(\theta) = \frac{\epsilon\left(\frac{1}{2}r^2\theta\right)}{d} = \frac{\epsilon r^2\theta}{2d} \tag{11}$$

where ϵ is the absolute permittivity of the dielectric medium.

Sensitivity (S) is formally defined as the differential rate of change of the output variable with respect to the input measured variable. Taking the derivative of capacitance with respect to angular displacement θ:

$$S = \frac{dC}{d\theta} = \frac{d}{d\theta}\left[\frac{\epsilon r^2\theta}{2d}\right] = \frac{\epsilon r^2}{2d} \tag{12}$$

Analyzing this final sensitivity expression block by block reveals three absolute truths: 1. Sensitivity is a strict constant; it is completely independent of the angular displacement θ itself (proving the device is perfectly linear). 2. Sensitivity is **directly proportional to the square of the radius** (r^2), not inversely proportional. 3. Sensitivity is **inversely proportional to the gap distance** (d), not directly proportional.

Orthogonal Diagram of Geometric Model:

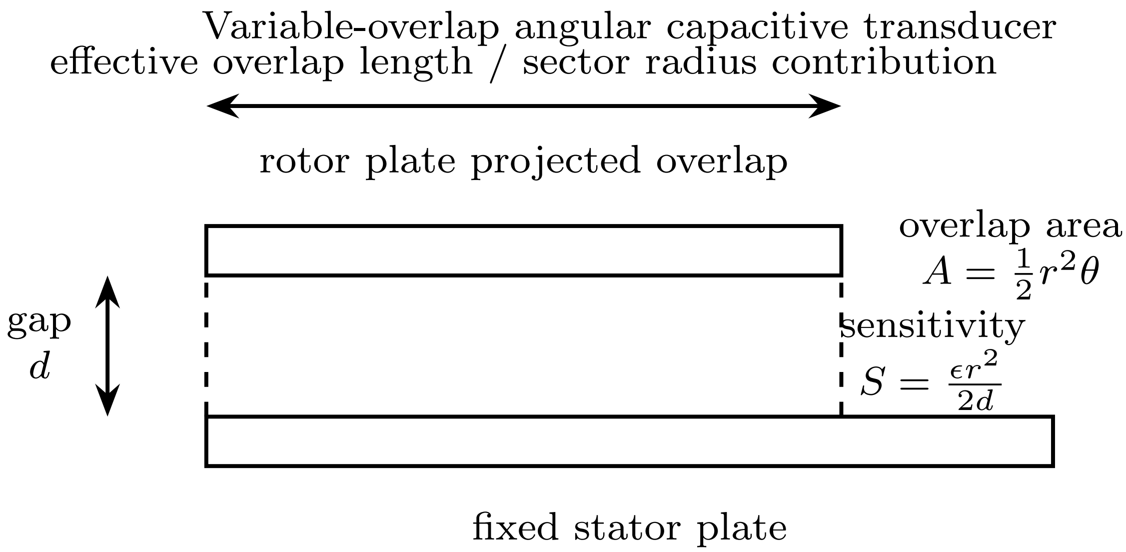

Figure 3: Orthogonal representation of a variable-overlap capacitive angular transducer.

1(d) LVDT Excitation and Maximum Measurable Frequency

Statement: "In an LVDT, the maximum measurable frequency of core motion is independent of the frequency of the excitation source used."

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Explanation: An LVDT operates on the rigorous principle of Amplitude Modulation (AM). The primary coil is driven by a high-frequency alternating current carrier wave:

$$E_{in}(t) = V_c \sin(\omega_c t) \tag{13}$$

where $f_c = \omega_c/2\pi$ is the carrier (excitation) frequency.

When the magnetic core moves, it acts as the modulating signal. Let the mechanical motion follow a dynamic trajectory:

$$x(t) = X_m \sin(\omega_m t) \tag{14}$$

where $f_m = \omega_m/2\pi$ is the mechanical frequency of the core motion.

The differential secondary output voltage is the continuous product of the mechanical position and the carrier signal:

$$E_{out}(t) = K \cdot x(t) \cdot E_{in}(t) = K \cdot X_m \sin(\omega_m t)\sin(\omega_c t) \tag{15}$$

Applying trigonometric expansion, this yields two distinct electrical sideband frequencies: $(f_c + f_m)$ and $(f_c - f_m)$.

To accurately recover the mechanical motion $x(t)$, the output must pass through a Phase-Sensitive Demodulator followed by a strict Low-Pass Filter. According to the Shannon-Nyquist theorem and practical filter design constraints for envelope detectors, the carrier frequency must act as an extremely fast sampling rate compared to the slow mechanical motion. If the mechanical frequency f_m approaches the carrier frequency f_c, the sidebands fold over the baseband signal, creating severe aliasing and phase distortion. The low-pass filter becomes physically incapable of separating the modulation envelope from the high-frequency carrier ripple.

Therefore, an absolute engineering constraint dictates that the excitation frequency must be significantly higher than the highest expected mechanical frequency. The accepted standard rule is:

$$f_c \geq 10 \times f_m \tag{16}$$

Consequently, the maximum measurable frequency of core motion ($f_{m,max}$) is strictly and fundamentally dependent upon, and limited by, the chosen excitation frequency (f_c).

Orthogonal Diagram of AM Signal Path:

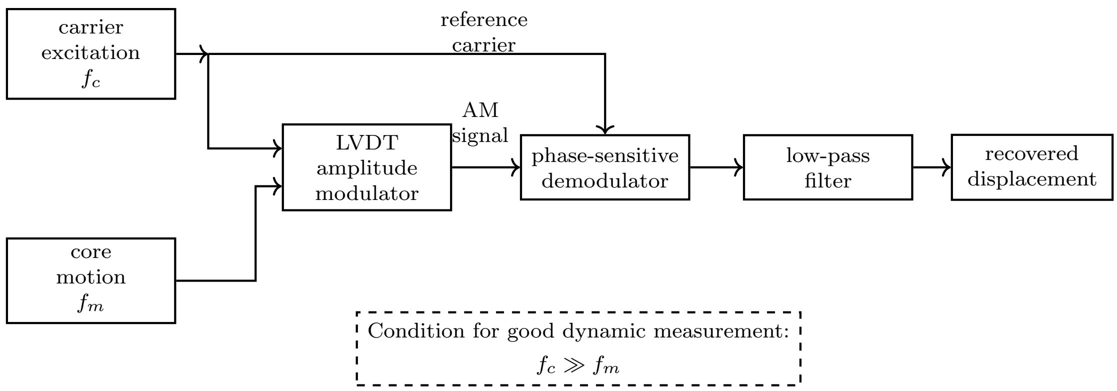

Figure 4: LVDT as an amplitude-modulation system with excitation frequency constraint.

1(e) Capacitive Liquid Level Gauge Sensitivity

Statement: "For a capacitive liquid level gauge, the sensitivity of the system is inversely proportional to the relative permittivity of the dielectric."

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Explanation: A standard capacitive liquid level gauge employs two concentric cylindrical metal electrodes submerged vertically into a tank. The fluid acts as a rising and falling dielectric medium. Let the inner cylinder radius be r_1 and the outer cylinder radius be r_2. The total height of the cylinders is L. The liquid level to be measured is h.

The system effectively operates as two cylindrical capacitors connected perfectly in parallel: 1. The lower section filled with the liquid (relative permittivity ϵ_{r1}). 2. The upper section filled with vapor or air (relative permittivity $\epsilon_{r2} \approx 1$).

The fundamental capacitance of a coaxial cylinder of length x is $C = \frac{2\pi\epsilon x}{\ln(r_2/r_1)}$. Summing the parallel liquid and vapor components yields the total dynamic capacitance:

$$C_{total} = C_{liquid} + C_{vapor} = \frac{2\pi\epsilon_0\epsilon_{r1}h}{\ln(r_2/r_1)} + \frac{2\pi\epsilon_0\epsilon_{r2}(L-h)}{\ln(r_2/r_1)} \tag{17}$$

Factoring out the common geometric terms:

$$C_{total} = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \left[\epsilon_{r1}h + \epsilon_{r2}L - \epsilon_{r2}h\right] \tag{18}$$

$$C_{total} = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \left[(\epsilon_{r1} - \epsilon_{r2})h + \epsilon_{r2}L\right] \tag{19}$$

The sensitivity of the gauge (S) is the rate of change of capacitance with respect to the physical liquid level height h:

$$S = \frac{dC_{total}}{dh} = \frac{d}{dh}\left(\frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \left[(\epsilon_{r1} - \epsilon_{r2})h + \epsilon_{r2}L\right]\right) \tag{20}$$

$$S = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)}(\epsilon_{r1} - \epsilon_{r2}) \tag{21}$$

By examining the final differential block, the sensitivity S sits firmly in the numerator. It is **directly proportional** to the difference in relative permittivities between the liquid and the vapor ($\epsilon_{r1} - \epsilon_{r2}$). It is absolutely not inversely proportional. The greater the permittivity of the liquid, the higher and more robust the sensitivity of the sensor.

Orthogonal Diagram of Coaxial Sensor Setup:

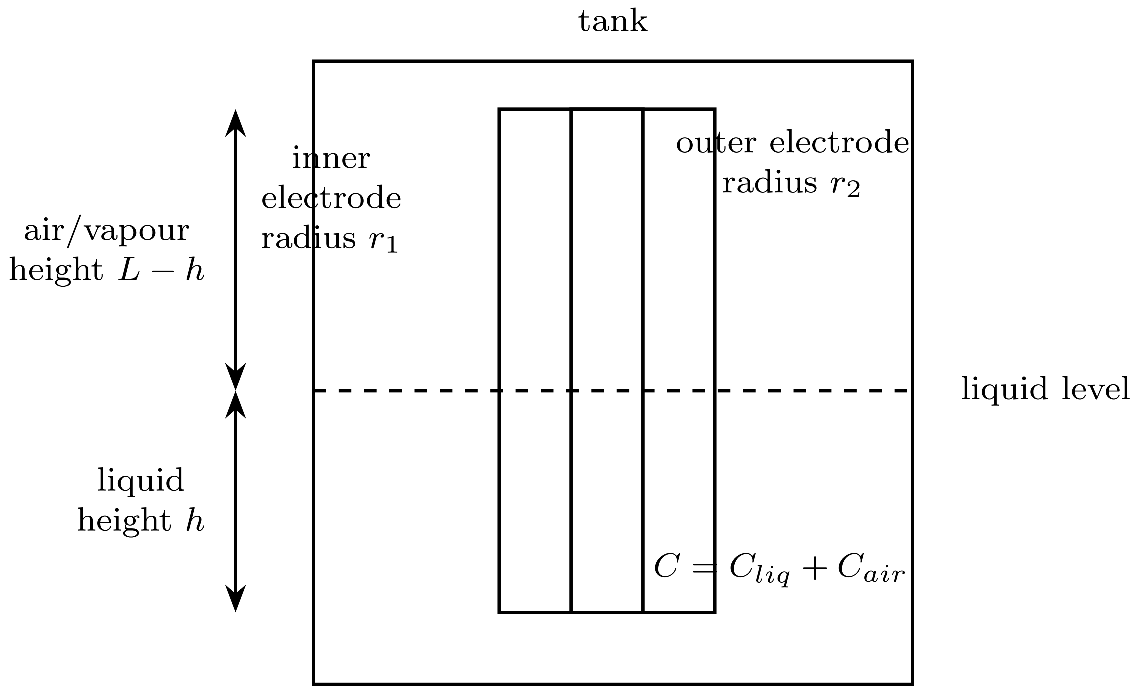

Figure 5: Coaxial capacitive liquid-level gauge with separated labels.

QUESTION 2: LVDT AND PIEZOELECTRIC PRINCIPLES

2(a) Residual Voltage in LVDT and Null Reduction Techniques

Concept of Residual Null Voltage: In a theoretically pristine Linear Variable Differential Transformer (LVDT), the secondary coils are wound identically and placed in perfect geometric symmetry relative to the primary coil. When the high-permeability magnetic core rests at the exact geometric center (the null position), the alternating magnetic fluxes linking secondary one (S_1) and secondary two (S_2) are absolutely identical. Because the secondaries are wired in strict series opposition, their induced voltages should cancel completely:

$$E_{out} = E_{s1} - E_{s2} = 0 \text{ Volts} \tag{22}$$

However, when tested with a highly sensitive oscilloscope, a continuous, non-zero alternating voltage is observed at the null position. This "residual voltage" destroys measurement resolution and introduces a physical dead-band. It stems from two grassroots physical imperfections:

1. **Harmonic Distortion (Non-Linearity):** The nickel-iron core exhibits a non-linear B-H magnetization curve. A pure sinusoidal primary excitation current produces a magnetic flux rich in odd harmonics (primarily the 3rd harmonic). While the fundamental frequency fluxes may cancel perfectly at the null center, the spatially distributed harmonic fluxes possess different wavelengths and phase relationships, preventing them from canceling simultaneously. 2. **Stray Capacitive Coupling (Quadrature Error):** Parasitic capacitance exists inherently between the high-voltage primary winding and the secondary windings. These microscopic capacitors allow stray alternating currents ($I = C\frac{dV}{dt}$) to leak directly into the secondary coils. Because this leakage current is purely capacitive, it generates error voltages that are shifted 90° out-of-phase (in quadrature) with the fundamental induced magnetic voltage. These quadrature signals do not cancel in series opposition.

Null Reduction Method (i): When a Centre-Tapped Voltage Source is Available
If the primary excitation source possesses a strict centre-tap, engineers can employ the **Wagner Earth** shielding technique. The exact centre of the primary voltage source is solidly tied to the true earth ground of the measurement system. This configuration perfectly balances the electric field potentials across the primary coil relative to ground. Consequently, any stray capacitive leakage currents from the upper and lower halves of

the primary coil are bypassed harmlessly directly to ground, rather than being forced through the high-impedance secondary windings. This eradicates the quadrature residual voltage component entirely.

Null Reduction Method (ii): When a Centre-Tapped Source is NOT Available
If an isolated floating source is used, the system cannot bypass the stray currents. Instead, the signal conditioning block must mathematical reject the error. The output signal is fed into a **Phase-Sensitive Demodulator (PSD)**. The PSD multiplies the LVDT output by the pristine primary reference voltage. If the residual error is a quadrature signal $E_{error}\sin(\omega_c t + 90°)$, multiplying it by the reference $V_{ref}\sin(\omega_c t)$ yields:

$$E_{mult} = E_{error}\cos(\omega_c t) \cdot V_{ref}\sin(\omega_c t) = \frac{E_{error}V_{ref}}{2}\sin(2\omega_c t) \tag{23}$$

When passed through a final Low-Pass Filter, the average DC value of the double-frequency sine wave $\sin(2\omega_c t)$ is strictly mathematical zero. The PSD naturally annihilates the residual null voltage without requiring physical grounding modifications.

Orthogonal Diagram of Stray Capacitance Origin:

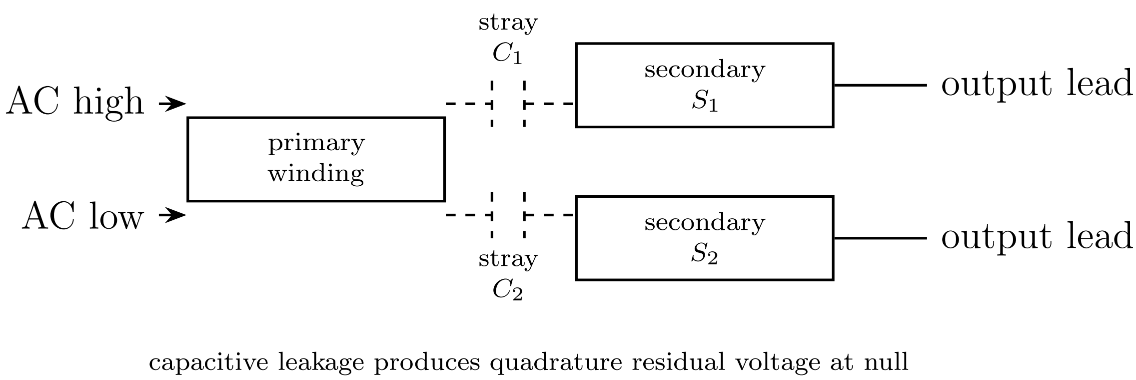

Figure 6: Origin of residual null voltage due to stray capacitances in LVDT.

2(b) Phase Sensitive Demodulation and Low Pass Filter

Why Phase Sensitive Demodulation (PSD) is Employed: An LVDT inherently generates an Amplitude Modulated (AM), suppressed-carrier alternating current signal. When the core moves to a position $+x$, the output is an AC wave of amplitude A. When the core moves to the exact opposite position $-x$, the output is an identical AC wave of amplitude A, but its phase is flipped by exactly 180° relative to the primary excitation.

If a conventional diode bridge rectifier (envelope detector) is utilized, it merely extracts the absolute mathematical magnitude $|A|$ of the AC signal. A conventional rectifier completely destroys the phase information. Therefore, the measurement system becomes blind to the physical direction of the core motion; +2 mm and −2 mm will yield the exact same positive DC voltage output.

A **Phase Sensitive Demodulator** solves this absolute failure. It acts as an electronic multiplier. It takes the LVDT secondary signal and continuously multiplies it by the pristine primary carrier reference signal. If the core is at $+x$ (in-phase):

$$E_{mult} = [Kx\sin(\omega_c t)] \times [V_{ref}\sin(\omega_c t)] = KxV_{ref}\sin^2(\omega_c t) \tag{24}$$

If the core is at $-x$ (out-of-phase by 180°):

$$E_{mult} = [-Kx\sin(\omega_c t)] \times [V_{ref}\sin(\omega_c t)] = -KxV_{ref}\sin^2(\omega_c t) \tag{25}$$

The polarity of the multiplied signal intrinsically retains the algebraic sign of the displacement.

Why a Low Pass Filter (LPF) is Employed: The output of the mathematical multiplier is not a clean DC voltage. Using the fundamental trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$, we expand the in-phase multiplier output:

$$E_{mult} = KxV_{ref}\left[\frac{1-\cos(2\omega_c t)}{2}\right] \tag{26}$$

$$E_{mult} = \frac{KxV_{ref}}{2} - \frac{KxV_{ref}}{2}\cos(2\omega_c t) \tag{27}$$

The block-by-block analysis shows the signal consists of two components: 1. A pure DC component: $\frac{KxV_{ref}}{2}$ (This contains the true spatial magnitude and direction). 2. A high-frequency AC ripple component oscillating at precisely twice the carrier frequency: $2\omega_c$.

To interface this sensor with an Analog-to-Digital Converter or a DC voltmeter, the high-frequency ripple must be violently suppressed. A strict **Low Pass Filter** is installed immediately following the multiplier. The cut-off frequency of this filter is engineered to be substantially lower than $2\omega_c$. The LPF strips away the $-\cos(2\omega_c t)$ term entirely, leaving only the pristine, proportional DC voltage that accurately represents both the span and the direction of the core's physical displacement.

Orthogonal Flow Diagram of PSD Signal Processing:

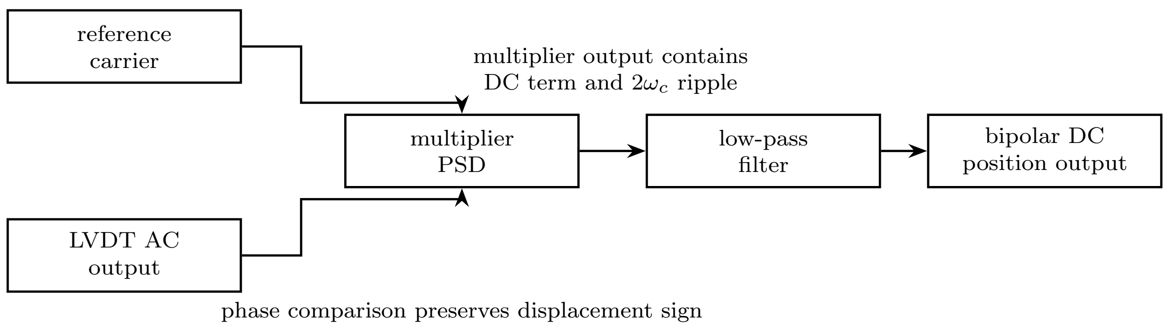

Figure 7: Phase-sensitive demodulation followed by low-pass filtering.

2(c) Piezoelectric ’g’ and ’d’ Constants Derivation

Definition of Fundamentals: Piezoelectric materials exhibit a coupled electro-mechanical behavior. Applying physical stress generates electrical charge, and applying electric fields generates mechanical strain.

1. **The ’d’ constant (Charge Sensitivity):** Defined as the total electrical charge (q) generated across the crystal surfaces strictly per unit of applied mechanical force (F). It dictates the fundamental generation capability of the specific crystal lattice.

$$d = \frac{q}{F} \quad \text{Units: } [Coulombs/Newton] \text{ or } [C/N] \tag{28}$$

2. **The ’g’ constant (Voltage Sensitivity):** Defined as the resulting electric field intensity (E) generated internally strictly per unit of applied mechanical stress (P).

$$g = \frac{E}{P} \quad \text{Units: } [Volt \cdot meters/Newton] \text{ or } [V \cdot m/N] \tag{29}$$

Derivation of the Relationship between ’g’ and ’d’: We must break down the ’g’ constant equation using fundamental electrostatics. Let the crystal have a total thickness t and a cross-sectional area A. The mechanical stress P is force per unit area:

$$P = \frac{F}{A} \tag{30}$$

The electric field intensity E across parallel surfaces is the voltage V divided by the thickness t:

$$E = \frac{V}{t} \tag{31}$$

Substituting these into the original ’g’ constant definition:

$$g = \frac{E}{P} = \frac{\left(\frac{V}{t}\right)}{\left(\frac{F}{A}\right)} = \frac{V \cdot A}{F \cdot t} \tag{32}$$

The crystal geometry mathematically forms a parallel-plate capacitor. The fundamental capacitance is $C = \frac{\epsilon A}{t}$, where ϵ is the absolute permittivity of the crystal material. From electrostatic charge laws, $V = \frac{q}{C}$. Substituting this definition of voltage:

$$g = \frac{\left(\frac{q}{C}\right) \cdot A}{F \cdot t} = \frac{q \cdot A}{C \cdot F \cdot t} \tag{33}$$

Now, substitute the capacitance formula $C = \frac{\epsilon A}{t}$ into the denominator:

$$g = \frac{q \cdot A}{\left(\frac{\epsilon A}{t}\right) \cdot F \cdot t} \tag{34}$$

The area A and thickness t terms physically cancel out entirely, leaving:

$$g = \frac{q}{\epsilon \cdot F} \tag{35}$$

Since we defined $d = \frac{q}{F}$, we arrive at the absolute block-by-block relation:

$$g = \frac{d}{\epsilon} \implies d = \epsilon \cdot g \tag{36}$$

Expressions for Charge Sensitivity and Voltage Sensitivity: If the transducer acts as a displacement sensor, the input variable is mechanical displacement x. The crystal acts as a highly stiff Hookean spring with stiffness constant K. Therefore, the applied force is $F = K \cdot x$.

1. **Charge Sensitivity (S_q)** in $[Coulombs/meter]$:

$$S_q = \frac{q}{x} = \frac{d \cdot F}{x} = \frac{d \cdot (K \cdot x)}{x} = d \cdot K \tag{37}$$

2. **Voltage Sensitivity (S_v)** in $[Volts/meter]$: Using $V = E \cdot t$, and substituting $E = g \cdot P = g \cdot \frac{F}{A}$:

$$V = g \cdot \left(\frac{K \cdot x}{A}\right) \cdot t \tag{38}$$

$$S_v = \frac{V}{x} = \frac{g \cdot K \cdot t}{A} \tag{39}$$

QUESTION 3: ADVANCED SENSOR CIRCUIT ANALYSIS

3(a) Capacitor Microphone Circuit and Frequency Response

Operating Principle and Setup: A capacitor microphone is a variable-gap capacitive transducer utilized for detecting high-frequency dynamic acoustic pressure waves (sound). The physical assembly consists of a rigid backplate and a highly flexible, ultra-thin metallic diaphragm stretched parallel to it. When sound waves strike the diaphragm, it dynamically deflects, causing the gap distance d to change over time, resulting in a time-varying capacitance $C(t)$.

To convert this varying capacitance into a measurable voltage, the sensor is placed in strict series with a high-voltage DC bias source (V_s) and a massive load resistance (R). The output voltage is extracted across the load resistance.

Derivation of the Transfer Function: By applying Kirchhoff's Voltage Law (KVL) around the single closed series loop, we sum the voltage drops:

$$V_s = V_R + V_C = i(t)R + \frac{1}{C(t)} \int i(t)dt \tag{40}$$

Because the DC source V_s is a rigid constant, taking the time derivative of the entire equation yields zero on the left side:

$$0 = R\frac{di(t)}{dt} + \frac{i(t)}{C(t)} + \left[\int i(t)dt\right] \frac{d}{dt}\left(\frac{1}{C(t)}\right) \tag{41}$$

Let the gap distance at rest be d_0. The dynamic displacement caused by the sound wave is $x_1(t)$. The distance becomes $d(t) = d_0 - x_1(t)$. The capacitance becomes:

$$C(t) = \frac{\epsilon A}{d_0 - x_1(t)} \approx C_0 \left(1 + \frac{x_1(t)}{d_0}\right) \tag{42}$$

where $C_0 = \frac{\epsilon A}{d_0}$ is the static resting capacitance.

In a grassroots perturbation analysis, assuming the dynamic displacement $x_1(t)$ is microscopically small compared to the resting gap d_0, the charge on the capacitor $q(t)$ remains approximately constant at its static bias level $Q_0 = V_s C_0$. Therefore, the current $i(t) = \frac{dq(t)}{dt}$ is governed by the change in capacitance. The output voltage is $e_0(t) = i(t)R$. By substituting the approximations into the differential equation and taking the Laplace transform, the complex differential equation collapses into a standard first-order high-pass transfer function:

$$\frac{E_0(s)}{X_1(s)} = \left(\frac{V_s}{d_0}\right)\left[\frac{s\tau}{1+s\tau}\right] \tag{43}$$

where the critical time constant is defined as $\tau = R \cdot C_0$.

Frequency Response and Range Enhancement: Substituting $s = j\omega$, the frequency response magnitude is:

$$|H(j\omega)| = \left(\frac{V_s}{d_0}\right)\frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \tag{44}$$

This is the exact mathematical signature of a **High-Pass Filter**. At high frequencies ($\omega\tau \gg 1$), the denominator approaches $\omega\tau$, and the gain flattens out to a constant sensitivity $S = \frac{V_s}{d_0}$. At low frequencies, the response plummets, causing severe attenuation and phase distortion. The lower cut-off frequency (the -3 dB point) where accurate measurement completely fails is:

$$f_{cut-off} = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC_0} \tag{45}$$

How to Enhance the Frequency Range: To accurately measure very deep, low-frequency bass sounds, the cut-off frequency $f_{cut-off}$ must be forced as close to zero as physically possible. This requires maximizing the time constant τ. 1. **Increase the Base Capacitance (C_0):** This is accomplished by manufacturing the gap d_0 as small as possible or increasing the diaphragm area A. However, this is bounded by physical breakdown limits and mechanical tension requirements. 2. **Increase the Series Resistance (R):** This is the primary engineering solution. Microphones employ staggering resistor values (often in the range of hundreds of Mega-Ohms, 10^8 Ω) coupled with Field Effect Transistor (FET) buffers to artificially drive the cut-off frequency below 20 Hz, ensuring full-spectrum acoustic fidelity.

Orthogonal Schematic of Capacitor Microphone:

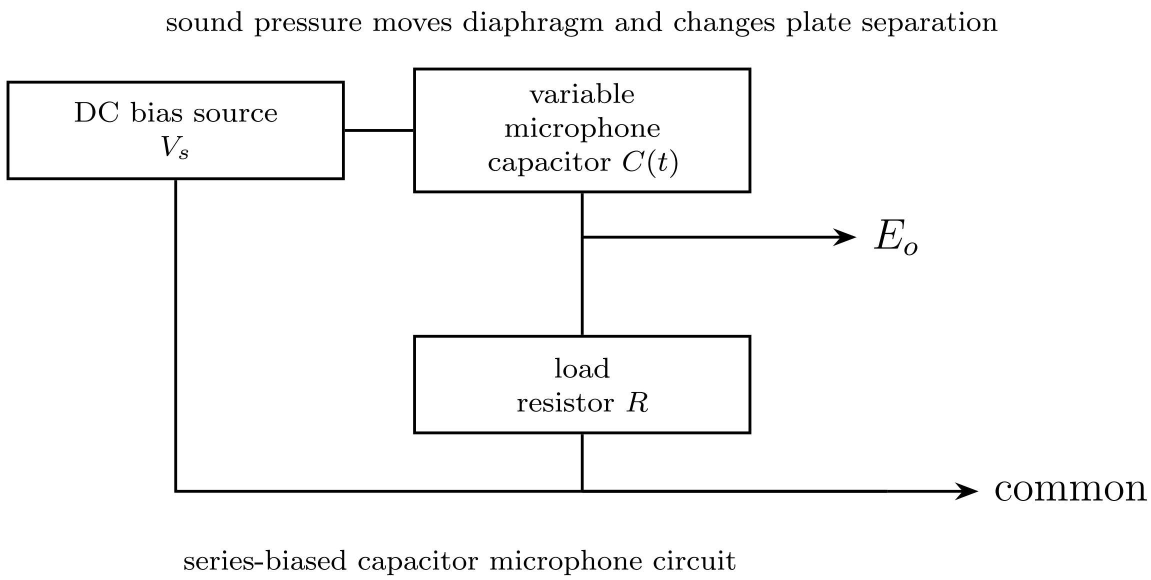

Figure 8: Capacitor microphone circuit with bias source and load resistor.

3(b) Piezoelectric System with Cable and Amplifier

System Construction: A complete practical piezoelectric transducer system cannot function as a standalone crystal. It requires three distinct physical blocks: 1. **The Piezoelectric Crystal:** Modeled electrically as a pure charge generator $q(t)$ in parallel with its inherent crystal capacitance C_p and a leakage resistance R_p. 2. **The Connecting Cable:** Coaxial cables possess significant capacitance between the central core and the outer shield. This is modeled as a massive shunt capacitor C_c placed in parallel with the crystal. 3. **The Amplifier (Voltage Mode):** The measurement device presents an input resistance R_a and an input capacitance C_a across the terminals.

Derivation of the Transfer Function: By utilizing Norton's equivalent circuit theorem, we combine all parallel elements into a single unified block. The total equivalent parallel capacitance is:

$$C_{eq} = C_p + C_c + C_a \tag{46}$$

The total equivalent parallel resistance is:

$$R_{eq} = \frac{R_p \cdot R_a}{R_p + R_a} \tag{47}$$

The charge generator produces a current $i_q = \frac{dq}{dt}$. In the Laplace domain, $I_q(s) = sQ(s)$. The total impedance of the parallel combination is:

$$Z(s) = \frac{R_{eq} \cdot \left(\frac{1}{sC_{eq}}\right)}{R_{eq} + \left(\frac{1}{sC_{eq}}\right)} = \frac{R_{eq}}{1 + sR_{eq}C_{eq}} \tag{48}$$

The output voltage $E_0(s)$ developed across this impedance is the product of current and impedance:

$$E_0(s) = I_q(s) \cdot Z(s) = sQ(s) \left[\frac{R_{eq}}{1 + sR_{eq}C_{eq}}\right] \tag{49}$$

Substitute the system time constant $\tau = R_{eq}C_{eq}$. Then $R_{eq} = \frac{\tau}{C_{eq}}$.

$$E_0(s) = Q(s) \left[\frac{s\left(\frac{\tau}{C_{eq}}\right)}{1 + s\tau}\right] = \frac{Q(s)}{C_{eq}} \left[\frac{s\tau}{1 + s\tau}\right] \tag{50}$$

If the system acts as a displacement sensor, $Q(s) = K_q X_i(s)$. The final transfer function is:

$$\frac{E_0(s)}{X_i(s)} = \left(\frac{K_q}{C_p + C_c + C_a}\right) \left[\frac{s\tau}{1 + s\tau}\right] \tag{51}$$

Measurement of Static Displacement: Can this system be used to measure a static (zero-frequency, DC) displacement? **No, it is absolutely physically impossible.** A static displacement means the input displacement $x(t)$ becomes a constant value, implying its frequency $\omega = 0$. Evaluating the magnitude of the high-pass transfer function at $\omega = 0$:

$$|G(j0)| = \left(\frac{K_q}{C_{eq}}\right) \frac{0 \cdot \tau}{\sqrt{1+0}} = 0 \text{ Volts} \tag{52}$$

Grassroots physics dictates that when a static force is applied and held, an initial charge $+q$ is generated instantly. However, because the total equivalent resistance R_{eq} is finite, it provides a closed electrical path for the charge to physically bleed away and neutralize over time. The output voltage decays exponentially to zero ($e^{-t/\tau}$) despite the heavy physical load still remaining clamped on the crystal. Therefore, piezoelectric systems are strictly dynamic-only instruments.

QUESTION 4: SHORT NOTES

4(a) Capacitive Transducer with Solid Dielectric of Variable Permittivity

Grassroots Principle: This transducer category does not operate by altering the gap distance (d) or the overlapping area (A). Instead, the entire physical geometry of the parallel plates remains completely locked and rigidly fixed. The measurement action occurs by physically sliding a solid slab of dielectric material (with relative permittivity $\epsilon_r > 1$) linearly between the plates. As the slab moves laterally, it replaces low-permittivity air ($\epsilon \approx 1$) with high-permittivity solid material. This profoundly alters the equivalent dielectric constant of the entire gap, causing the total capacitance to rise smoothly.

Block-by-Block Mathematical Derivation: Let the parallel plates have a total length L, a fixed width w, and a constant separation gap d. Let a solid dielectric slab of matching width w and thickness d (perfectly filling the vertical gap without friction) be inserted laterally by a distance x.

The system automatically fractures into two distinct, mathematically parallel capacitors side-by-side: 1. **The Air-Filled Region:** Length $(L - x)$.

$$C_{air} = \frac{\epsilon_0 \cdot \epsilon_{air} \cdot w \cdot (L - x)}{d} \approx \frac{\epsilon_0 \cdot w \cdot (L - x)}{d} \tag{53}$$

2. **The Dielectric-Filled Region:** Length x.

$$C_{dielectric} = \frac{\epsilon_0 \cdot \epsilon_r \cdot w \cdot x}{d} \tag{54}$$

Because they are physically adjacent sharing the same massive electrode plates, they add in parallel. The total capacitance $C(x)$ is:

$$C(x) = C_{air} + C_{dielectric} = \frac{\epsilon_0 w(L - x)}{d} + \frac{\epsilon_0 \epsilon_r w x}{d} \tag{55}$$

$$C(x) = \frac{\epsilon_0 w L}{d} + \frac{\epsilon_0 w x(\epsilon_r - 1)}{d} \tag{56}$$

The first term is the initial static capacitance with zero insertion ($C_{initial}$). The sensitivity (S) of this transducer is strictly constant:

$$S = \frac{dC}{dx} = \frac{\epsilon_0 w(\epsilon_r - 1)}{d} \tag{57}$$

Because the sensitivity derivative yields a strict constant devoid of any variables, this confirms the transducer provides a **perfectly linear** output response over its entire operational range.

Orthogonal Diagram of Variable Dielectric:

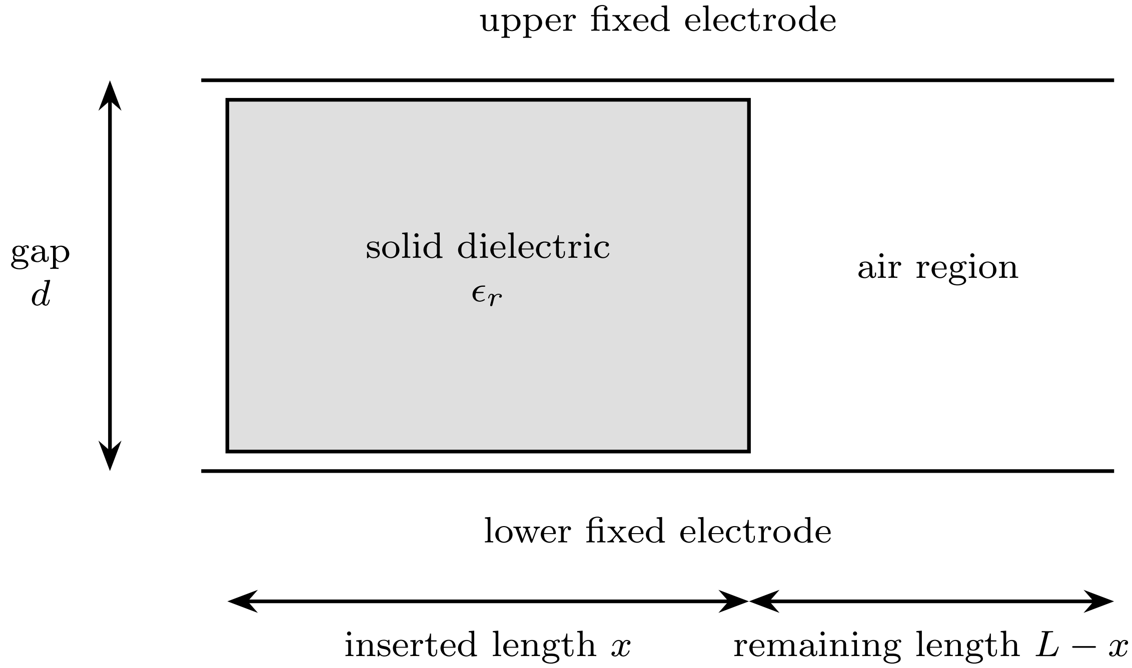

Figure 9: Capacitive transducer using a sliding solid dielectric slab.

4(b) Synchronous Demodulation Technique Employed in LVDT

The Grassroots Necessity: An LVDT output is heavily Amplitude Modulated. Displacement $+x$ and $-x$ produce the exact identical output voltage magnitude, differing only by a stark 180° phase reversal. A simple diode rectifier acts purely as an absolute value function ($|V_{out}|$), destroying the phase shift data and causing massive ambiguity regarding which side of the null center the core resides.

Synchronous demodulation (Phase-Sensitive Demodulation) resolves this by using the primary excitation oscillator not just as power, but as a rigid timing reference for mathematical multiplication.

Block-by-Block Execution: The demodulator is built utilizing a precision four-quadrant analog multiplier or a synchronous switching ring-diode bridge. Let the primary reference voltage be $V_r(t) = V_c \sin(\omega_c t)$. Let the LVDT output be $V_s(t) = K \cdot x \cdot \sin(\omega_c t + \phi)$, where ϕ is strictly $0°$ (for $+x$) or $180°$ (for $-x$).

The synchronous multiplier forcefully multiplies these two waveforms in real-time:

$$V_{mult}(t) = [K \cdot x \cdot \sin(\omega_c t + \phi)] \times [V_c \sin(\omega_c t)] \tag{58}$$

Applying trigonometric product-to-sum identities:

$$V_{mult}(t) = \frac{K \cdot x \cdot V_c}{2}[\cos(\phi) - \cos(2\omega_c t + \phi)] \tag{59}$$

This composite signal is immediately routed into a steep Low-Pass Filter block to annihilate the high-frequency $2\omega_c$ harmonic ripple. The final integrated DC output is:

$$V_{DC} = \frac{K \cdot x \cdot V_c}{2} \cos(\phi) \tag{60}$$

Analyzing the Output Phase Integrity: 1. If the core is pulled right ($+x$), $\phi = 0°$, making $\cos(0°) = +1$. The DC output becomes a strong positive voltage ($+V$). 2. If the core is pushed left ($-x$), $\phi = 180°$, making $\cos(180°) = -1$. The DC output mathematically flips to a strong negative voltage ($-V$). Furthermore, this technique inherently possesses massive noise immunity. Any stray background noise oscillating at a random non-synchronous frequency ω_{noise} will average out to absolute zero after multiplication and low-pass filtering.

Orthogonal Switch Demodulator Topology:

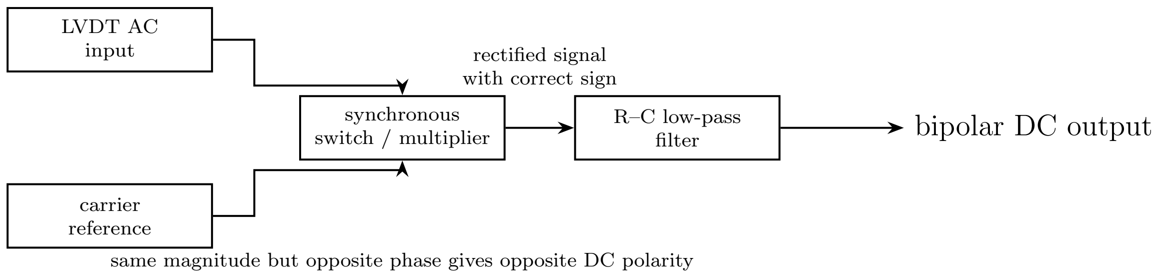

Figure 10: Synchronous demodulator used with an LVDT.

4(c) Piezoelectric Accelerometers

Grassroots Mechanical Concept: A piezoelectric accelerometer does not measure acceleration by sensing space and time. It employs Newton's Second Law of Motion ($F = m \cdot a$) in a rigid mechanical housing. A known, calibrated internal seismic mass (M) rests directly atop a stiff piezoelectric crystal block. The entire assembly is mechanically pre-loaded under heavy compression by a heavy-duty spring. This ensures the mass never physically separates from the crystal during violent upward decelerations, as piezoelectric ceramics fracture easily under tension.

When the rigid outer housing experiences an external acceleration ($a = \ddot{x}$), inertia dictates that the internal mass resists the change in velocity. This exerts a dynamic compressive force $F_{dynamic} = M \cdot a$ down onto the crystal structure.

Electro-Mechanical Conversion: The piezoelectric crystal possesses a charge sensitivity constant d (Coulombs/Newton). Upon being crushed by the dynamic force, the crystal lattice deforms, displacing charge centers and producing a localized surface charge:

$$q = d \cdot F_{dynamic} = d \cdot M \cdot a \tag{61}$$

This mathematically proves that the electrical charge generated is perfectly directly proportional to the applied acceleration.

Because the crystal acts as a highly rigid spring (high stiffness K), the natural resonant frequency of the sensor ($\omega_n = \sqrt{K/M}$) is exceptionally high (often in the tens of kiloHertz). This grants the sensor a spectacularly wide flat-frequency bandwidth, making it the supreme choice for analyzing violent high-frequency machinery vibrations and ballistic shock impacts.

Electrical Interfacing Constraint: Because a static continuous acceleration (like constant gravity) produces a static charge that rapidly bleeds away through internal leakage resistance, this accelerometer cannot measure static states. Furthermore, to read the minuscule generated picocoulombs of charge without suffering catastrophic attenuation from coax cable capacitance, the sensor must be strictly mated to a high-impedance **Charge Amplifier**. The charge amplifier utilizes a feedback capacitor (C_f) to perfectly convert the charge packet into a rugged, low-impedance voltage signal ($V = q/C_f$).

Orthogonal Diagram of Accelerometer Core:

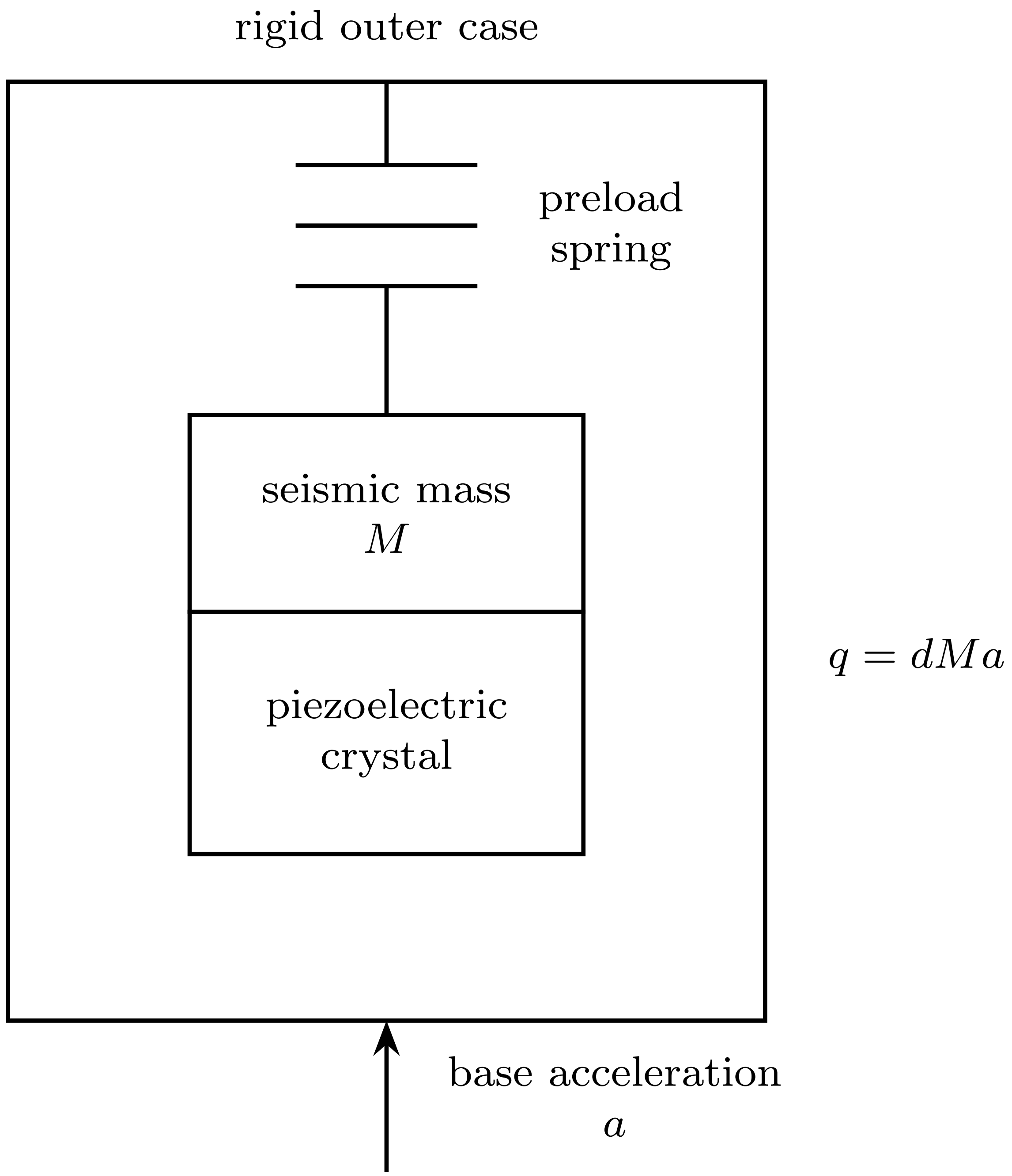

Figure 11: Piezoelectric accelerometer using seismic mass and piezoelectric crystal.

QUESTION 5: ADVANCED BRIDGE AND SYSTEM CONSTRAINTS

5(a) Transformer Ratio Bridges for Capacitance Measurement

The Engineering Problem (Why): Standard AC Wheatstone and Schering bridges fail catastrophically when attempting to measure microscopic capacitances (in the picofarad range) typical of capacitive transducers. The fundamental flaw lies in **stray capacitance to ground**. Parasitic capacitances exist between the bridge corner nodes and the metal chassis. Because these stray capacitances are of similar magnitude to the tiny sensor capacitance, they completely dominate the bridge balance equation, introducing massive errors that cannot be easily calibrated out without extraordinarily complex shielding (Wagner Earth setups).

The Transformation Solution (How): A Transformer Ratio Bridge abandons resistive ratio arms entirely. Instead, it employs a highly precision-wound, tightly magnetically coupled transformer with a strictly center-tapped secondary coil.

The transformer generates two perfectly opposite AC voltages, V_1 and V_2, dictated purely by the turns ratio of the upper coil (N_1) to the lower coil (N_2).

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \tag{62}$$

The unknown transducer capacitance (C_x) is placed in one arm, and a calibrated precision standard capacitor (C_s) is placed in the other. A null detector is placed at the center junction. At the exact null balance point, the net current entering the detector node is absolute zero. Therefore, the admittance equations must balance:

$$V_1 \cdot (j\omega C_x) = V_2 \cdot (j\omega C_s) \tag{63}$$

Substituting the strict turns ratio:

$$C_x = C_s \left(\frac{V_2}{V_1}\right) = C_s \left(\frac{N_2}{N_1}\right) \tag{64}$$

Why This is Immune to Stray Capacitance: The true genius of the transformer ratio bridge lies in its immunity to grounding errors. Any stray capacitance present at the output node of the transformer is driven by an ideal voltage source possessing virtually zero internal impedance. The transformer forcefully dictates the voltage ratio regardless of how heavily the node is loaded by parasitic strays. The turns ratio N_2/N_1 is locked rigidly by physical physics (the number of copper turns on an iron core), allowing measurement precision to exceed 1 part in 10^6.

Orthogonal Diagram of Transformer Ratio Bridge:

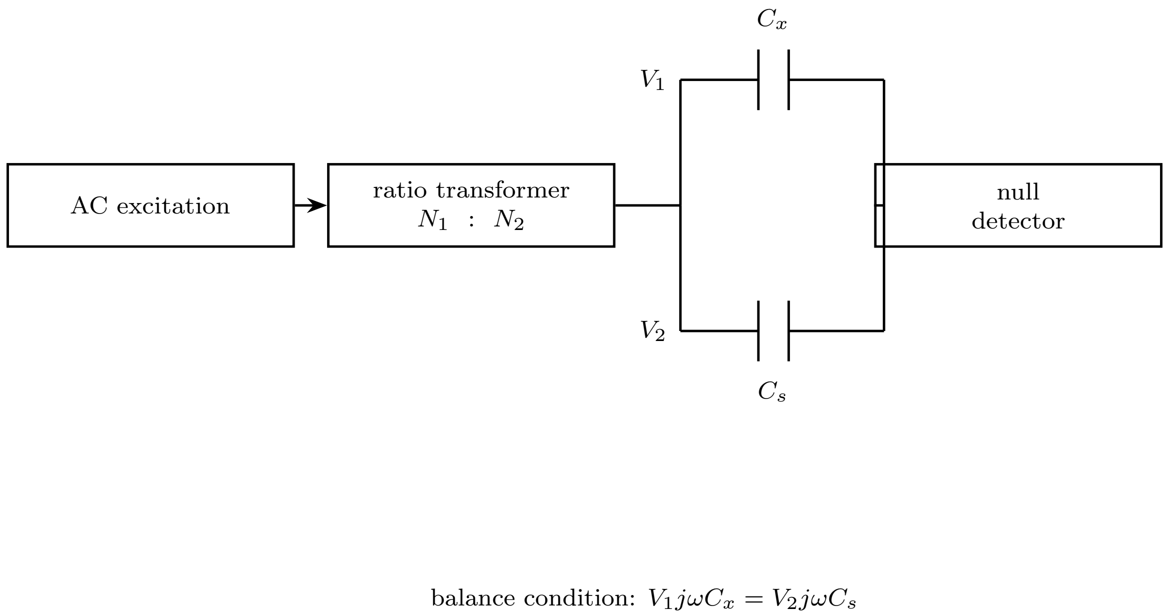

Figure 12: Transformer ratio bridge for capacitance measurement.

5(b) Practical Charge Amplifier Feedback Constraint

Statement: "A practical charge amplifier circuit can be designed with only a capacitor in the feedback path."

Correction & Justification: The statement is **Incorrect**.

Grassroots Analytical Failure of the "Only Capacitor" Design: A theoretical, ideal charge amplifier utilizes an operational amplifier with an infinite input impedance and a purely capacitive feedback loop (C_f). It behaves as a perfect mathematical integrator. The charge q generated by the piezoelectric sensor is forcefully swept into C_f by the virtual ground, yielding a perfect output $E_0 = -q/C_f$.

However, in harsh, real-world practical engineering, Operational Amplifiers are physically imperfect. They possess intrinsic, unavoidable input bias currents (I_{bias}) flowing into the inverting and non-inverting terminals. If the feedback path contains **only** a capacitor C_f, this bias current has absolutely no DC resistive path to escape to ground. Consequently, the microscopic bias current continuously forces charge onto the feedback capacitor ($V_c = \frac{1}{C_f} \int I_{bias} dt$). The voltage across the capacitor ramps up linearly as a function of time. Within fractions of a second, the output voltage slams violently into the positive or negative supply rails of the op-amp (e.g., +15V or −15V).

Once the amplifier hits saturation, it is completely paralyzed and fundamentally blind to any AC signals coming from the transducer.

The Mandatory Practical Solution: To forcefully prevent this catastrophic DC saturation, a practical charge amplifier absolutely must be designed with a massive stabilization **resistor** (R_f) installed strictly in parallel with the feedback capacitor (C_f). This high-value resistor (typically 10^9 to 10^{11} Ohms) provides a continuous, vital DC leakage path for the op-amp's bias currents, effectively bleeding them away and anchoring the op-amp's quiescent DC operating point stably at zero volts. While this unfortunately transforms the perfect integrator into a high-pass filter, dictating a low-frequency cut-off limit ($\omega_c = 1/R_f C_f$), it is the only physical method to maintain amplifier stability.

Orthogonal Diagram of Practical Charge Amplifier:

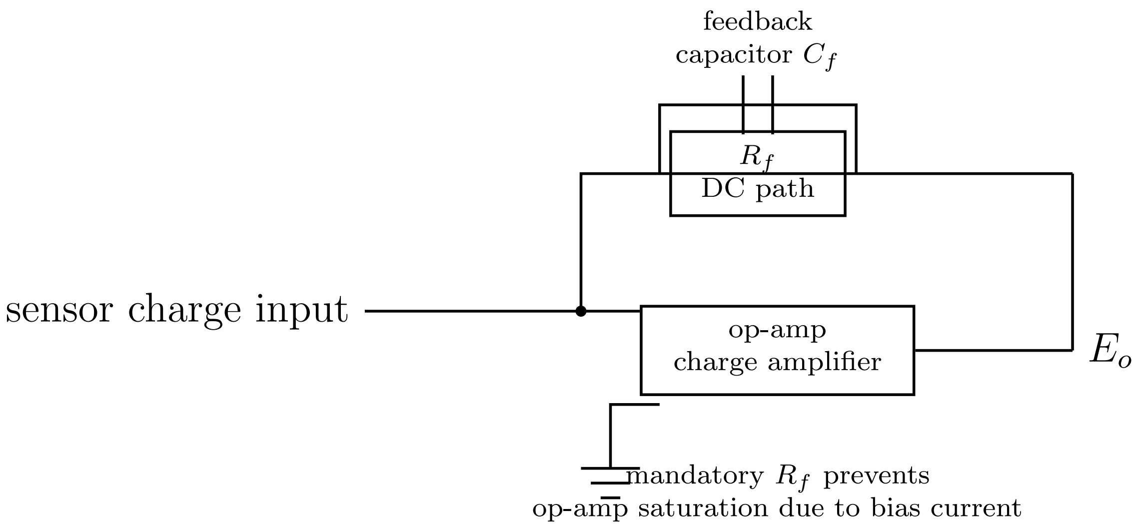

Figure 13: Practical charge amplifier with parallel R_f and C_f feedback.

5(c) LVDT Direction and Magnitude Spectrum

Statement: "The direction of core-motion of an LVDT is reflected in the magnitude spectrum of its output voltage."

Correction & Justification: The statement is **False**.

Grassroots Analytical Proof: An LVDT acts as a precision spatial-to-electrical Amplitude Modulator. When the high-permeability core is displaced by a physical distance $+x_0$ (e.g., pushed into the right secondary coil), the instantaneous output AC voltage waveform is:

$$V_{out,right}(t) = K \cdot x_0 \cdot \sin(\omega_c t) \tag{65}$$

When the core is displaced by the exact same physical distance $-x_0$ (pushed into the left secondary coil), the instantaneous output AC voltage waveform is flipped structurally:

$$V_{out,left}(t) = K \cdot (-x_0) \cdot \sin(\omega_c t) = -K \cdot x_0 \cdot \sin(\omega_c t) \tag{66}$$

Using standard trigonometric identities, the negative sign is mathematically equivalent to a pure 180° phase shift. Thus, the equation rewrites exactly to:

$$V_{out,left}(t) = K \cdot x_0 \cdot \sin(\omega_c t + 180^\circ) \tag{67}$$

To compute the magnitude spectrum, we perform the absolute mathematical Fourier Transform of both time-domain signals. For the right displacement ($+x_0$), the frequency domain magnitude strictly at the carrier frequency ω_c is:

$$|V_{out,right}(j\omega_c)| = K \cdot x_0 \tag{68}$$

For the left displacement ($-x_0$), the magnitude is calculated as the square root of the sum of the squares of the real and imaginary components. The phase shift only affects the argument (angle), not the scalar magnitude vector length. Therefore:

$$|V_{out,left}(j\omega_c)| = K \cdot x_0 \tag{69}$$

Both spectra yield an identical magnitude peak of exactly $K \cdot x_0$ at the exact same carrier frequency ω_c. If an engineer feeds the LVDT output into a standard Spectrum Analyzer (which predominantly displays amplitude magnitude versus frequency), the $+x_0$ graph and the $-x_0$ graph will be indistinguishable. They will overlay perfectly on top of each other. The magnitude spectrum completely destroys and ignores all vector directionality. The strict direction of core motion is embedded exclusively within the **Phase Spectrum** (which jumps discontinuously between 0° and 180° depending on the core's position relative to the null center point).